

# Introduction

In this notebook we are analyzing the market of data-field freelancing so we can find useful or informative insights. In this analysis we are looking at UpWork & Guru data-field freelancers, in terms of skills, earnings etc ...  
So let's look at we have

**NOTE:** Data-field refers to tech jobs that work mainly with data which are:

- Data analysts
- Data engineers
- Data scientests
- Machine learning developers

Things we are analysing in this notebook:

- [Q1](#). Earnings distribution
- [Q2](#). Common skills
- [Q3](#). Hour rate change
- [Q4](#). Average job cost

## Setting up

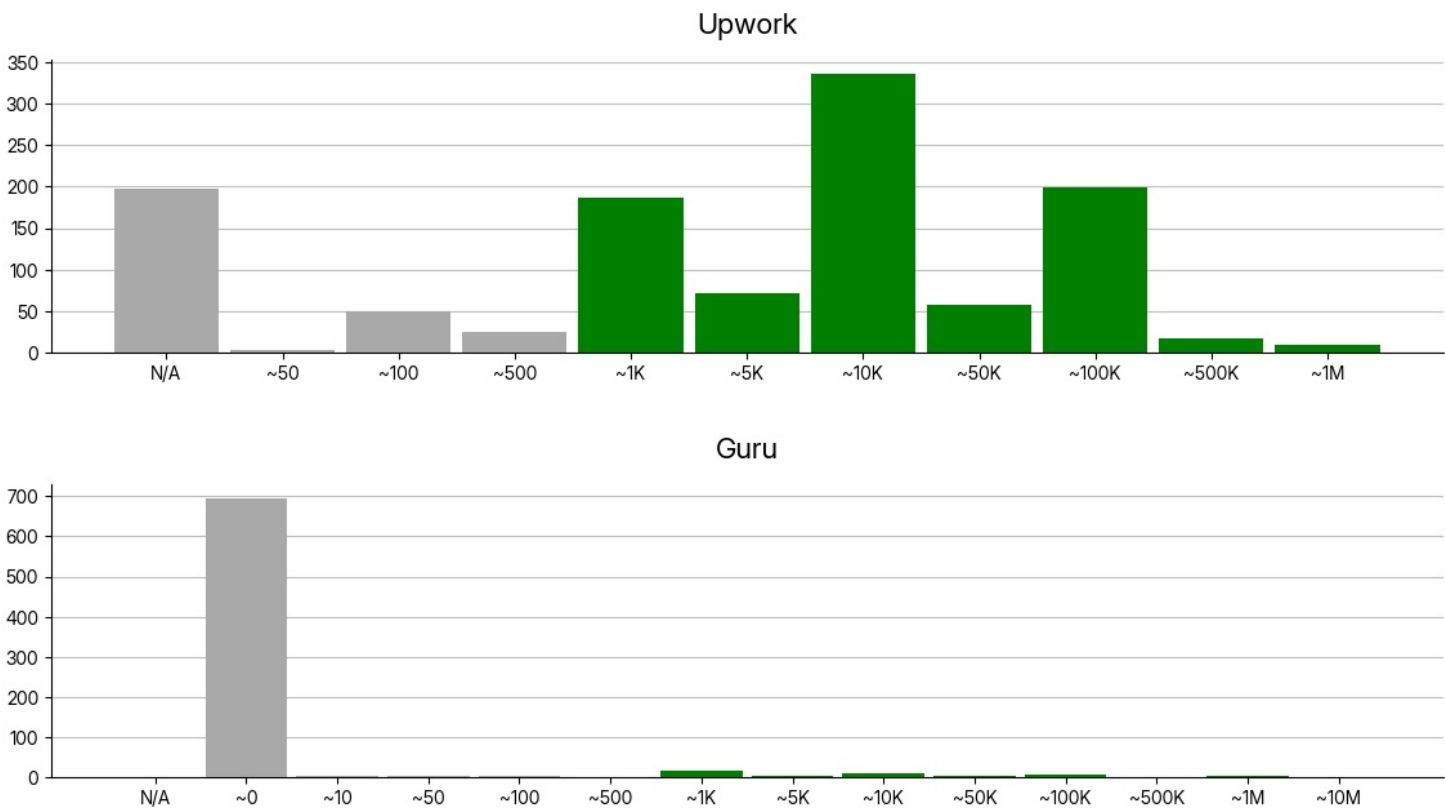
## Analysis

### Q1. Earnings distribution

First thing I want to see is earnings distribution in Guru and UpWork so we can see if it's possible to achieve any success on any of both and how possible it is.

Though we won't be able to see how quick it was -to earn money- because we don't have access to the freelancers' account age.

Data-Field Freelancers total earnings distribution



Oh, firstly we can confidently say that it's really really rare to earn anything on Guru. There are only few people who acieved a lot but more than 95% of the people didn't earn anything, which means I will not be conducting further analysis to Guru's data.

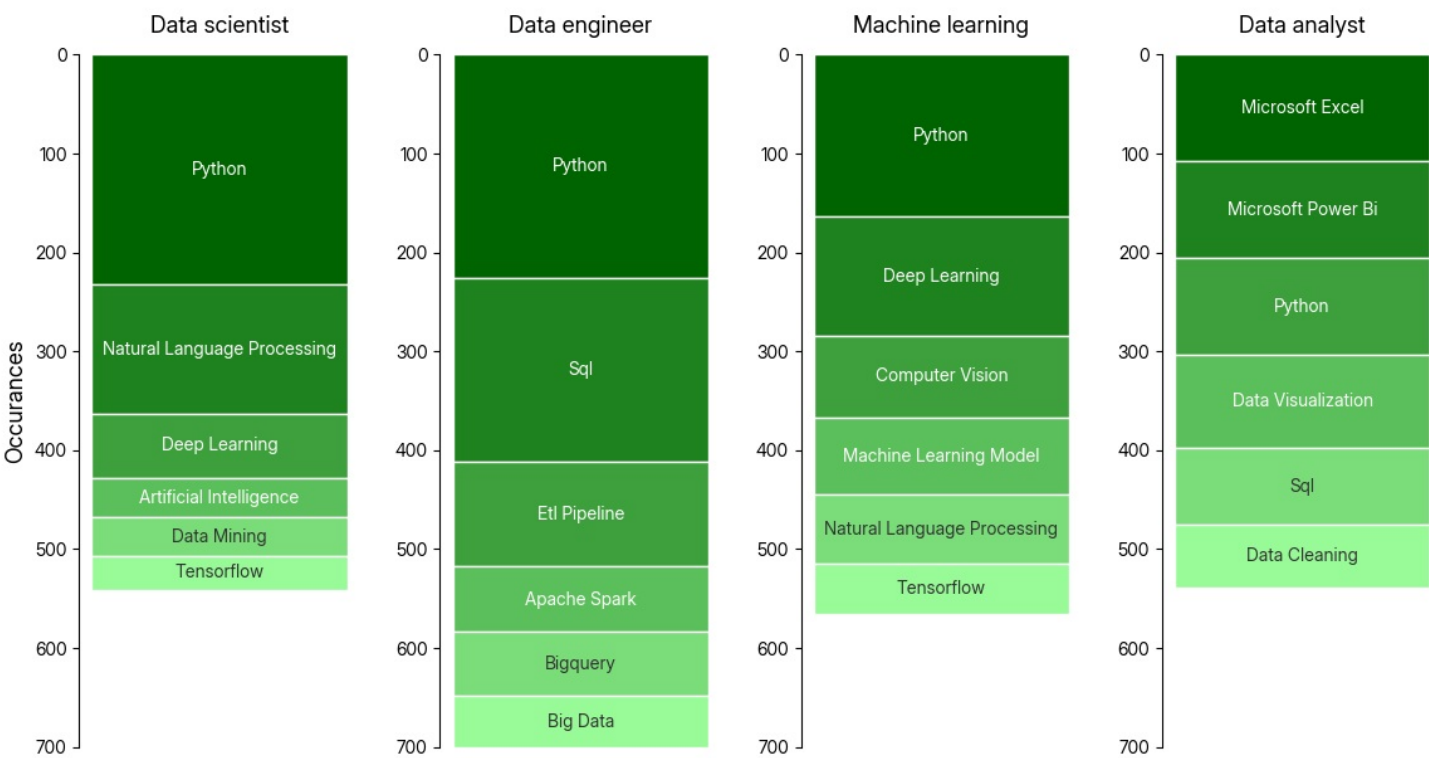
Also we can see how common it is in UpWork to have earned some amount of freelancing though we can't be quite sure of how fast not easy was it to achieve those amounts but at least we can see decent earnings.

So at least we can say that you can earn decent amounts on UpWork.

## Q2. Common skills

Now we should take a look at what are the essential skills freelancers need in thier profiles for each data-field job title.

Most common skills in UpWork freelancers profiles



Interesting, Python is the most common skills among data-field freelancer's but we can also that "Data analyst" freelancers use Excel more often than Python (*with small difference*).

Also if you've read my jobs analysis notebook (Q4) you can see that Python is common here as it was there, but one notable difference is how ununiform skills are here.

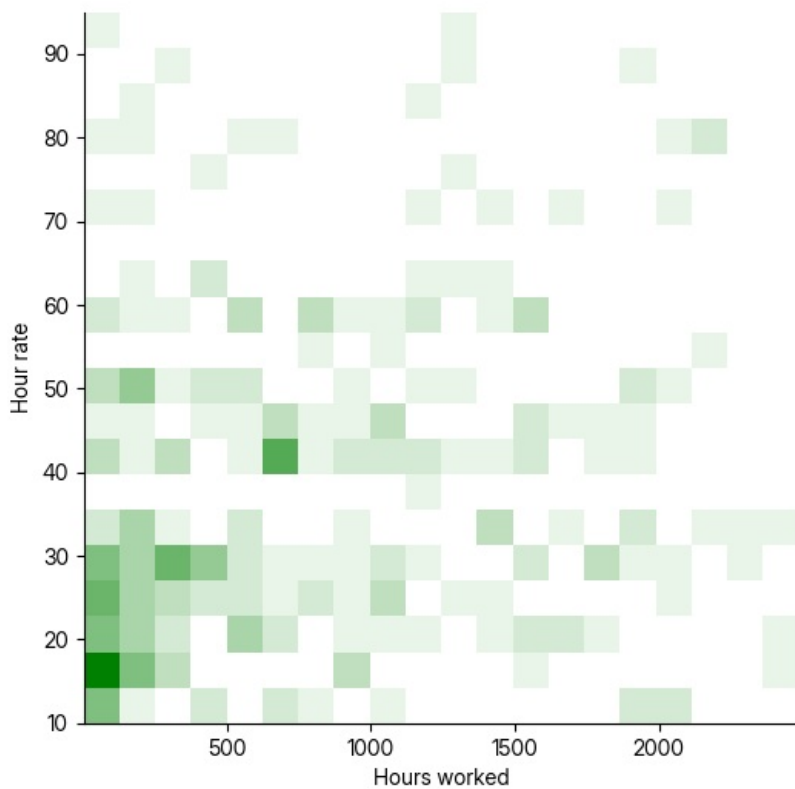
## Q3. Hour rate change

Now let us take a look at the correlation between how many hours did the freelancer work and how much does he price his hour of work.

**NOTE:** This might not be the most useful chart because of how many factors might play into this.

## 'Hour rate' & 'Total hours worked' correlation in UpWork

\*Fixed priced jobs done doesn't count in the 'Hours worked'



Now let's take a look at the [Pearson's correlation coefficient](#) to see the correlation in a mathematical way

And also we should take a look at the [P value](#) to see how significant is "Hours worked"

correlation and significance between 'Hours worked' and 'Hour rate'

Pearson's  $r = 0.1872526663408876$   
P value =  $0.001963174564893831$

Simply this means that there's a positive relation between 'Hours worked' and 'Hour rate' and it's not just some random noise because  $P < 0.05$

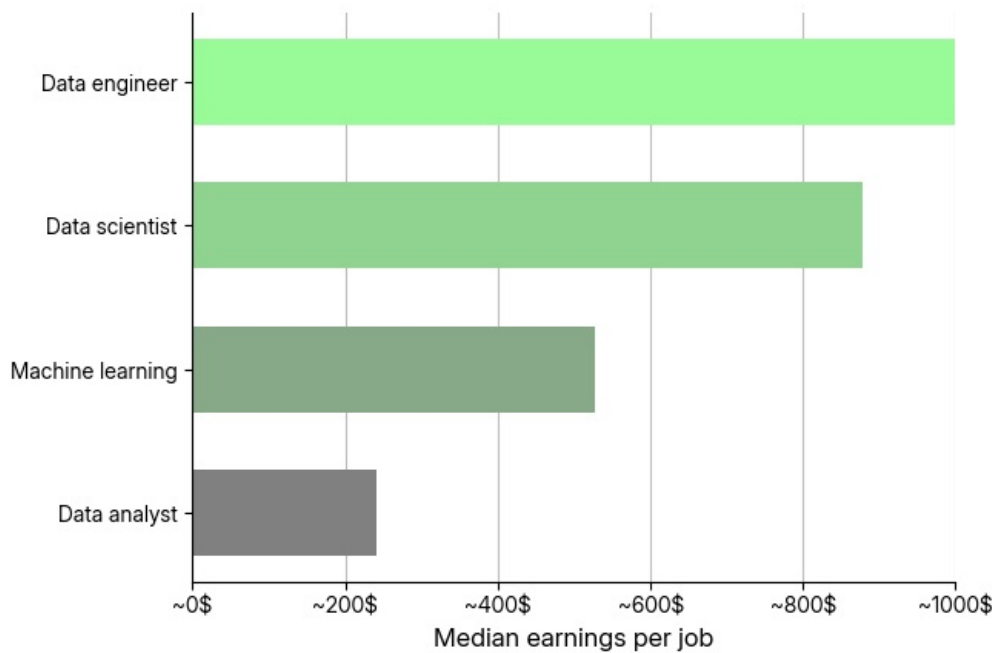
Though the correlation is not that strong.

So now we can conclude that the more hours a freelancer has worked the more likely they are to increase their pay rates

## Q4. Average job cost

Finally I want to find how much does every job title of the freelancers earn per job done e.g. *How much does a data scientist make per order (job)?*

## Avg 'earnings per job' for each job title in UpWork



I actually expected that data engineers will have the highest 'Earnings per job'. But what's surprising is that Machine learnings devs get half what the data engineers get per job but keep in mind that we don't know how much work do they (freelancers) do per job or how much time does it take.

## Summary

Thanks for looking into my humble analysis for the data-field analysis market hope you found it informative here's a summary of all of the findings:

**NOTE:** This summary was the only thing in the notebook written with an LLM

- **Q1. Earnings Distribution:**

Guru: Over 95% of freelancers earned nothing, indicating extremely low chances of success. Further analysis was discontinued for Guru.

UpWork: Earnings are more common and generally decent, showing that freelancers can realistically achieve financial success on this platform (although the speed of earning is unknown).

- **Q2. Common Skills:**

Python is the most common skill among data-field freelancers overall. Excel slightly surpasses Python specifically for data analysts.

- **Q3. Hourly Rate vs. Hours Worked**

There is a positive correlation between hours worked and hourly rate. The relationship is statistically significant but not very strong, meaning freelancers who work more hours tend to charge higher rates, but many other factors are also involved.

- **Q4. Average Earnings per Job**

Data engineers earn the highest amount per job. Machine learning freelancers earn about half as much per job as data engineers. time or effort per job is unknown, so earnings per job should not be interpreted as earnings per hour.